

Skills Summary

Differentiation Rules Reference

Use this reference while completing your worksheet or when studying.

Choose the rule before you differentiate. A sum of powers: power rule term by term. Two non-constant factors multiplied: product rule. One expression divided by another: quotient rule. A constant over a power (like $5/x^3$) is *not* a quotient-rule problem — rewrite it as $5x^{-3}$.

Skill 1 — The Power Rule and Rewriting First

Rule: for every real n ,

$$\frac{d}{dx} x^n = nx^{n-1},$$

and constants differentiate to 0. Rewrite before you start: c/x^n is cx^{-n} , and $\sqrt{x}$ is $x^{1/2}$. The exponent goes down by one even when it is negative or fractional — x^{-3} becomes $-3x^{-4}$.

Example: Differentiate $y = 3x^4 - 7x + 2$

Step 1. Every term is already a power of x , so the power rule applies term by term — no product or quotient rule is involved.

Step 2. $3(4)x^3 = 12x^3$, $-7x^1$ gives -7 , and the constant 2 gives 0.
 $\Rightarrow y' = 12x^3 - 7$

Try it: Differentiate $y = 5x^3 + 2/x$

check: $15x^2 - 2x^{-2}$

Skill 2 — The Product Rule

Rule:

$$(uv)' = u'v + uv'$$

Derivative of the first times the second, plus the first times the derivative of the second. It is a *sum*, not a product: $(uv)' \neq u'v'$. Label u and v in the margin before differentiating, then expand and collect.

Example: Differentiate $y = (x + 3)(x^2 - 1)$

Step 1. Two non-constant factors are multiplied, so this is the product rule — take $u = x + 3$ and $v = x^2 - 1$.

Step 2. $u'v + uv' = (1)(x^2 - 1) + (x + 3)(2x) = x^2 - 1 + 2x^2 + 6x = 3x^2 + 6x - 1$.
 $\Rightarrow y' = 3x^2 + 6x - 1$

Try it: Differentiate $y = (2x - 1)(x^2 + 4)$

check: $8 + 2x - 2x^2$

Skill 3 — The Quotient Rule

Rule:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

The order in the numerator matters — swapping the two terms flips every sign in the answer. Leave the denominator as v^2 and simplify only the numerator.

Example: Differentiate $y = (x + 1)/(x - 2)$

Step 1. One expression is divided by another and neither is constant, so use the quotient rule with $u = x + 1$ and $v = x - 2$.

Step 2. $\frac{(1)(x-2)-(x+1)(1)}{(x-2)^2} = \frac{x-2-x-1}{(x-2)^2} = \frac{-3}{(x-2)^2}$.
 $\Rightarrow y' = \frac{-3}{(x-2)^2}$

Try it: Differentiate $y = 3x/(x + 4)$

check: $\frac{3(x+4)-x(3)}{(x+4)^2}$

(!) **Watch out:** the two commonest slips are $u'v'$ for a product and u'/v' for a quotient. Both are wrong, and both give an answer that looks tidier than the right one — tidiness is not evidence.